

The effect of insufficient quadriceps strength on gait after anterior cruciate ligament reconstruction

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Abstract

Objective. To determine the effect of quadriceps strength and joint stability on gait patterns after anterior cruciate ligament injury and reconstruction.

Design. Cross-sectional comparative study in which four groups underwent motion analysis with surface electromyography.

Background. Individuals following anterior cruciate ligament rupture often demonstrate reduced knee angles and moments during the early stance phase of gait. Alterations in gait can neither be ascribed to instability nor to quadriceps weakness alone when both are present.

Methods. Twenty-eight individuals with complete anterior cruciate ligament rupture (10 patients with acute rupture, 8 patients following reconstruction with quadriceps strength >90% of the uninvolved side [strong-anterior cruciate ligament reconstructed group], and 10 patients after reconstruction with quadriceps strength <80% of the uninvolved side [weak-anterior cruciate ligament reconstructed group]), and 10 uninjured subjects underwent an examination of their lower extremity to collect kinematics, kinetics, and electromyography during walking and jogging. Anterior cruciate ligament reconstruction was arthroscopically assisted and a double loop semitendinosus-gracilis autograft or allograft was used as a graft source. All reconstructed subjects had stable knees, full range of motion, and no effusion or pain at the time of testing (more than three months after surgery).

Results. Knee angles and moments of the strong group were indistinguishable from the uninjured group during early stance of both walking and jogging. The weak subjects had reduced knee angles and moments during walking, and jogged similarly to the deficient subjects. Regression analysis revealed a significant effect between early stance phase knee angles and moments and quadriceps strength during both walking and jogging.

Conclusion. Inadequate quadriceps strength contributes to altered gait patterns following anterior cruciate ligament reconstruction.

Relevance

Rapid strengthening following anterior cruciate ligament injury or reconstruction may contribute to a safe return to high-level activities. © 2002 Elsevier Science Ltd. All rights reserved.

Keywords: Quadriceps; Anterior cruciate ligament; Gait; Knee; Quadriceps avoidance

1. Introduction

Injury to the anterior cruciate ligament (ACL) is common amongst both regular participants in sports and other active individuals, and precipitates excessive anterior tibio-femoral laxity [1]. An inability to control this laxity results in joint instability and is usually manifested as an episode of giving way, a common complaint

among individuals who are ACL deficient [2–4]. Research has shown that the amount of joint laxity is not associated with a patient's instability following an ACL rupture, so the terms laxity and instability are not synonymous [5,6]. Many patients require reconstruction of the ACL to reduce the symptoms of instability and allow a return to high levels of activity [1].

Individuals who experience instability with ACL rupture do so when the injured limb accepts the weight of the body in early stance. During weight acceptance, the limb accepts full support of the body and attenuates shock through knee flexion that is controlled by an

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eccentric contraction of the quadriceps. When people experience knee instability, they alter their movement patterns at this point in the gait cycle through reduced knee flexion and reduced internal knee extensor moments [7,8]. These gait alterations may result from the unwillingness to fully activate the quadriceps muscles at a range close to full knee extension. Berchuck and Andriacchi [7] termed the reduced knee extensor moment “quadriceps avoidance” because near full extension, a quadriceps contraction may lead to anterior tibial translation [9,10] that could initiate an episode of giving way. The reduced knee moment, however, could represent either a reduction in quadriceps activation, or an increase in flexor activity provided by the hamstrings [2], or the gastrocnemius [11] and EMG data must be used to fully understand the implications of a reduced knee extensor moment. Altered movement patterns persist following anterior cruciate ligament reconstruction [12,13] despite surgical restoration of knee stability. This suggests that instability is not the sole cause of the gait adaptations.

While the literature provides some evidence that altered joint angles and moments are compensations for knee instability, it is conceivable that they are also the result of quadriceps femoris strength deficits. Quadriceps strength deficits are ubiquitous following both ACL rupture [14–16], and ACL reconstruction [17–19], and have been correlated with movement characteristics during both walking [11,20] and jogging [11,21]. This indicates that both instability and quadriceps strength deficits could be contributing to the gait deviations seen in these patient populations.

ACL rupture has been associated with early degenerative changes to the articular cartilage of the knee [1,22]. The instability that often accompanies an ACL rupture may contribute to the development of articular cartilage defects [23]. Degenerative changes also occur after reconstructive surgery, despite the elimination of the patient’s instability [1,22,23], which suggests that instability is not the sole cause of the degenerative changes. Quadriceps muscle weakness, which is prevalent in both ACL deficient and reconstructed patients, has recently been suggested to contribute to knee osteoarthritis [24]. If quadriceps femoris weakness contributes to altered movements and muscular adaptations, then reduced quadriceps strength might also contribute to the increased risk of arthritic changes in patient populations following ACL rupture or reconstruction. In addition, many of the gait deviations occur during weight acceptance, when an eccentric quadriceps contraction is necessary to control knee flexion and provide shock absorption. So insufficient quadriceps strength could reduce the ability of the knee to attenuate shock, which could lead to degenerative changes.

The purpose of this study was to investigate the effect of quadriceps femoris strength and instability on gait in

persons with anterior cruciate ligament rupture who undergo operative stabilization. Those who have quadriceps weakness following surgery were expected to exhibit diminished knee angles and reduced internal knee extensor moments during weight acceptance, similar to the adaptation that ACL deficient subjects exhibit following injury [8]. Subjects with nearly normal quadriceps strength following reconstruction were expected to resume movement characteristics that are similar to uninjured subjects.

2. Methods

2.1. Subjects

Twenty-eight individuals who had a complete isolated tear of the anterior cruciate ligament and experienced instability participated in this study. Instability was objectively identified through a screening process developed by Fitzgerald et al. [4]. Ten of these 28 subjects (six males, four females) were tested while still ACL deficient (ACL-D group) and were all less than 6 months from the time of the injury. The other 18 individuals had undergone a unilateral, endoscopic, arthroscopically assisted ACL reconstruction by one of two surgeons using a graft that did not violate the extensor mechanism (i.e. allograft or double loop semitendinosus-gracilis autograft). All subjects had knee instability prior to reconstruction, but had clinically stable knees at the time of testing (e.g. no buckling, shifting, or episodes of giving way, normal Lachman and pivot shift tests). These 18 subjects were divided into two groups based on their quadriceps strength. Eight subjects (five males, three females) comprised the strong-anterior cruciate ligament reconstructed (ACLR) group, which was operationally defined as those with quadriceps strength of 90% or more in the involved limb compared to the uninvolved quadriceps. Ten subjects (four males, six females) comprised the weak-ACLR group which was operationally defined as those with 80% or less than that of the uninvolved quadriceps (see Table 1). In a normal population, less than a 10% side-to-side difference in quadriceps strength can be expected, however, greater than a 20% difference is considered abnormal [25]. All subjects in the weak-ACLR group were tested at a minimum of 12 weeks following surgery to allow for ample time to recover quadriceps femoris strength. All subjects in the ACL-D group and six individuals in each of the ACL reconstructed groups were regular participants in activities involving jumping, pivoting, hard cutting, or lateral movements prior to injury. The remaining subjects were involved in either lower level activities (jogging, running, or swimming) or were sedentary. Ten uninjured subjects (eight males, two females), with no history of knee injury, were also tested

Table 1
Subject demographics

	Strong-ACLR	Weak-ACLR	ACL-D	Uninjured
Age (years)	21.4 (6.0) [15–31]	25.0 (7.8) [14–42]	28.1 (8.8) [16–43]	32.2 (6.6) [23–41]
Time since surgery (weeks)	14.3 (2.4) [9–17]	20.8 (9.4) [13–43]		
Quadriceps index (%)	95.3 (5.0) ^a [90.4–103.9]	67.6 (10.4) [47.5–80.4]	75.3 (11.3) [62.4–97.3]	
KOS-ADLS (%)	90.7 (11.1) [69–100]	92.0 (6.0) [85–100]		

Values indicate: mean (SD) [range].

^a The strong-ACLR group had a higher quadriceps index than both the weak-ACLR and ACL-D groups ($P = 0.000$).

as a control group. All subjects were able to extend their knees to at least zero degrees of extension, and had sustained no damage to the contralateral knee. No subject had knee pain or joint effusion at the time of testing. Patients with identified articular cartilage defects or concomitant injury to other knee ligaments were excluded from the study. Subjects were included in the study in order as they presented to our clinic, provided they met all inclusion criteria. All subjects were informed of the purpose of the study, and signed an informed consent form, approved by the Human Subjects Review Board of the University of Delaware, prior to testing. All patients completed the Activities of Daily Living Scale of the Knee Outcome Survey (KOS-ADLS), a self-report of the function on the subject's knee to determine their level of knee instability [26].

2.2. Strength testing

Quadriceps strength was tested as described previously [16,27]. Briefly, each subject was seated on an isokinetic dynamometer (KinCom, Chattanooga Group, Chattanooga, TN, USA) and produced a four second maximum voluntary isometric contraction with their knee flexed to 90°. During this contraction, electrodes (ConMed, Utica, NY, USA) placed over the quadriceps femoris muscle produced a supramaximal burst of electrical current to fully stimulate the quadriceps. If the subject failed to generate volitional force within 95% of the electrically elicited force, the test was repeated, up to four times with 5 min rest between tests. All subjects were able to produce maximal force on both sides within one session and therefore had no reflex inhibition. Both legs were tested and the quadriceps index, a measure of relative strength, was calculated (quadriceps index = involved volitional force/uninvolved volitional force $\times$ 100).

2.3. Motion and electromyographic testing

Electromyographic (EMG) data were collected at 960 Hz using active surface electrodes with an eight-channel

hard-wired EMG system (Motion Lab Systems, Baton Rouge, LA, USA) for the weak-ACLR and strong-ACLR subjects, and an eight-channel telemetry EMG system (B&L Engineering, Santa Fe Springs, CA, USA) for the ACL-D and uninjured subjects. Elastic bands (SuperWrap™, FabriFoam, Exton, PA, USA) were wrapped over the electrodes to secure them bilaterally to the muscle bellies of the medial head of the gastrocnemius, and mid-muscle belly of the lateral hamstrings and vastus lateralis.

Motion testing was done with a six camera, passive, three-dimensional motion analysis system (VICON, Oxford Metrics, London, UK) recording at 120 Hz. Calibration errors were kept below 3 mm. Force data were recorded from a six-component force plate (Bertec, Worthington, OH, USA) sampling at 960 Hz. The cameras detected retro-reflective markers affixed to the subject's shoes and placed in triads on rigid shells on the posterior aspect of the pelvis, thigh, and shank. Anatomical markers were placed over the head of the fifth metatarsal, lateral malleolus, lateral femoral condyle, greater trochanters, and iliac crests. Motion, force, and EMG data were collected concurrently as subjects performed six trials of free-speed walking and jogging along a 30-foot runway. Velocity data were monitored with two photoelectric beams to ensure that velocity varied not more than 5%. Trials were used if the subject remained at a consistent velocity and landed with their foot in the center of the force plate without targeting.

2.4. Data management and analysis

Walking and jogging kinematic data were low-pass filtered at 6 and 10 Hz, respectively. Force plate data were low-pass filtered at 50 Hz. The marker clusters, attached to the subject's pelvis, thighs, shanks, and feet, were used to track each segment's three-dimensional coordinate system. Move3D software (Move3D, NIH Biomechanics Laboratory, Bethesda, MD, USA) was used to calculate the knee's sagittal plane motions.

Joint kinetics were calculated using inverse dynamics, and are expressed as net internal moments. Knee angles and moments were normalized to 100% of the stance phase of gait. Joint moments were normalized to the subject's body mass multiplied by the leg length. The subject's six trials were averaged to obtain a mean for each subject.

The strong-ACLR and weak-ACLR subjects' raw EMG data were band-pass filtered between 20 and 350 Hz. Data from the ACL-D and uninjured group's muscles, collected with the telemetric system, were high-pass filtered with an eighth-order Butterworth filter at 90 Hz. For all groups, a linear envelope was created for analysis using full-wave rectification, followed by a second-order, phase corrected low-pass Butterworth filter with a cut-off frequency of 20 Hz. The linear envelope was normalized to the maximum muscle activity recorded during either a dynamic or maximum voluntary isometric contraction trial. Custom-designed software (Labview, National Instruments, Austin, TX, USA) was used to calculate the integral of the muscle activity from 100 ms prior to initial contact (to account for electro-mechanical delay [28]) until peak knee flexion. All EMG data were compared to the subject's uninvolved side. The differences in collection equipment necessitated the need for different filtering parameters, and therefore made it impossible to compare between groups.

Group differences were determined using one-way analyses of variance for the knee angle and moment at both initial contact and peak knee flexion. When differences were detected, *t*-tests were used as post hoc tests with a Bonferroni adjustment used to account for the multiple comparisons performed; therefore, significance was set at $p < 0.017$. A MANOVA with two repeated measures (muscle and side) was used to analyze the EMG integral data within each group, however no group comparisons were made because of the use of different EMG data collection systems. Regression analyses were used to determine the relative effect of quadriceps strength and muscle activity on the knee angles and moments at peak knee flexion.

3. Results

There was no difference between the weak-ACLR and strong-ACLR groups with respect to mean age ($P = 0.296$), mean time from surgery to testing ($P = 0.076$), or mean KOS-ADLS scores ($P = 0.594$) (Table 1). The strong-ACLR group's quadriceps index of 95.3% was significantly greater than the weak-ACLR group's quadriceps index of 67.6% ($P = 0.000$), and the ACL-D group's 75.3% ($P = 0.000$). There was no significant difference between the quadriceps indices of the weak-ACLR and ACL-D groups ($P = 0.140$).

4. Walking

There was no significant difference in the mean velocity between groups ($P = 0.089$). The weak-ACLR and strong-ACLR subjects walked at 1.98 m/s/leg length. The ACL-D group maintained 2.07 m/s/leg length, while the uninjured subjects walked at 2.22 m/s/leg length.

4.1. Kinematics and kinetics

The weak-ACLR, strong-ACLR, and ACL-D groups' knee angles at initial contact were not significantly different from those of the uninjured group ($P = 0.729$) (Table 2). At peak knee flexion, the weak-ACLR group showed a trend towards a diminished knee angle ($P = 0.043$) (Fig. 1) and had a significantly reduced knee moment ($P = 0.001$) (Fig. 2) compared to the uninjured group. The ACL-D group had a trend toward a decreased internal knee moment at peak knee flexion ($P = 0.021$).

The ACL reconstructed subjects' quadriceps index accounted for a significant portion of the variance in knee angle ($P = 0.036$; $R^2 = 0.260$) at peak knee flexion. Quadriceps index and the score on the KOS-ADLS together accounted for a significant portion of the variance in the ACL reconstructed subjects' knee moment at peak knee flexion ($P = 0.010$; $R^2 = 0.604$).

4.2. Muscle activity

There were no significant side-to-side differences in EMG integrals for the medial gastrocnemius, vastus lateralis, and lateral hamstrings for all patient groups.

5. Jogging

The weak-ACLR and strong-ACLR groups jogged at 3.36 and 3.48 m/s/leg length, respectively, during testing. The ACL-D group jogged at 3.33 m/s/leg length, while the uninjured group maintained 3.75 m/s/leg length. There was no significant difference in the jogging speed between groups ($P = 0.485$).

5.1. Kinematics and kinetics

At initial contact, the patient groups' knee angle was not significantly different from that of the uninjured group ($P = 0.275$) (Fig. 3). Both the weak-ACLR ($P = 0.056$) and ACL-D ($P = 0.063$) groups showed a trend towards a reduced knee flexion angle at peak knee flexion. The internal knee extension moments of the weak-ACLR ($P = 0.003$), and ACL-D ($P = 0.014$) groups were significantly reduced compared to those of the uninjured subjects at peak knee flexion (Fig. 4). The

Table 2

Gait characteristics for groups during walking and jogging

	Walking				Jogging			
	Weak- ACLR	Strong- ACLR	ACL-D	Uninjured	Weak- ACLR	Strong- ACLR	ACL-D	Uninjured
<i>Knee angle</i>								
Initial contact (degrees)	5.94 (3.12)	6.00 (3.99)	5.47 (6.11)	7.78 (5.29)	12.09 (4.20)	13.45 (3.93)	12.25 (6.07)	16.34 (6.53)
Peak knee flexion (degrees)	20.99 ^a (6.20)	24.10 (3.93)	21.86 (7.01)	26.54 (4.87)	38.52 (7.37)	41.29 (4.13)	38.98 (6.76)	46.12 (9.18)
<i>Knee moment</i>								
Peak knee flexion (%BW × LL)	0.25 ^b (0.26)	0.41 (0.33)	0.39 ^c (0.30)	0.71 (0.26)	1.28 ^d (0.72)	1.82 (0.64)	1.49 ^e (0.73)	2.34 (0.67)

Values indicate mean (SD).

^a $P = 0.043$.^b $P = 0.001$.^c $P = 0.021$.^d $P = 0.003$.^e $P = 0.014$.

strong-ACLR group, however, was not different from that of the uninjured group in any kinematic or kinetic variable we tested (Table 2).

There was a significant relationship between the quadriceps index and the knee angle ($P = 0.033$; $R^2 = 0.254$) and moment ($P = 0.006$; $R^2 = 0.379$) at peak knee flexion in the weak-ACLR and strong-ACLR subjects. The inclusion of the KOS-ADLS did not strengthen the relationship for the reconstructed subjects as a group. Separation into the weak-ACLR and strong-ACLR groups, however, revealed that for the strong-ACLR group, quadriceps index and the KOS-ADLS explained 88% of the variability in the knee moment at peak knee flexion ($P = 0.015$; $R^2 = 0.878$). The KOS-ADLS and quadriceps index were not able to explain the variability in the knee moment at peak knee flexion for the weak-ACLR group. ($P = 0.221$; $R^2 = 0.530$).

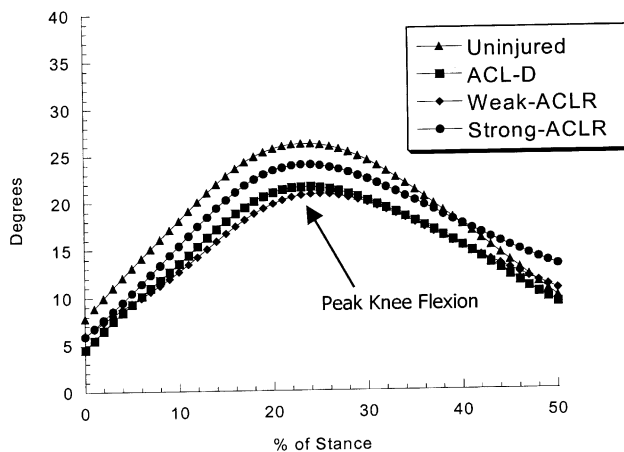


Fig. 1. Knee flexion angles during the weight acceptance phase of stance during walking.

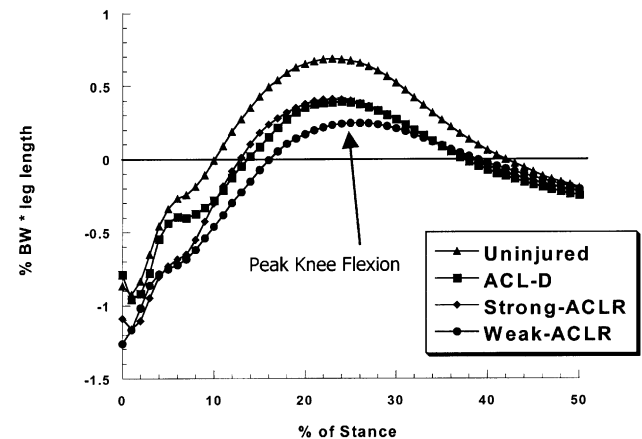


Fig. 2. Net internal knee moments during early stance phase of walking, normalized to body mass and leg length (N/m/kg m). A positive value represents a greater relative contribution from the knee extensors (i.e. quadriceps femoris).

5.2. Muscle activity

The magnitudes of muscle activity in the vastus lateralis and gastrocnemius muscles were not different in the involved and uninvolved sides of the weak-ACLR, strong-ACLR, and ACL-D groups. The lateral hamstrings, however, were significantly more active in the involved limb of the ACL-D group ($P = 0.029$) (Fig. 5).

6. Discussion

The hypothesis that quadriceps strength deficits contribute to the reduced knee angles and moments during early stance in ACL deficient and reconstructed subjects was supported by the data, however, the muscle

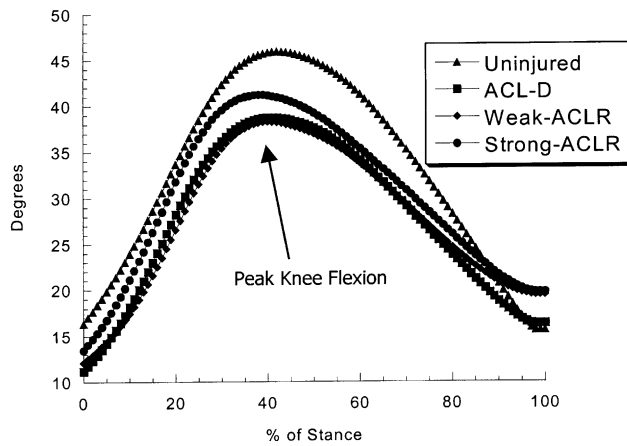


Fig. 3. Knee flexion angles during jogging.

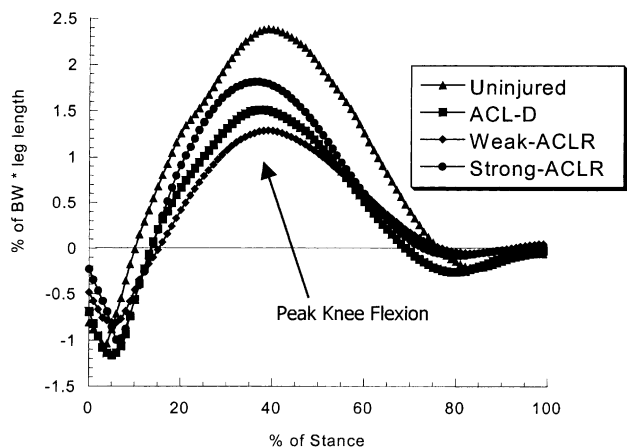


Fig. 4. Net internal knee moments during jogging, normalized to body mass and leg length (N m/kg m). A positive value represents a greater relative contribution from the knee extensors (i.e. quadriceps femoris).

activation patterns used to compensate for the altered movements are different. Only the strong group walked and jogged similarly to the uninjured group. Those who had quadriceps weakness following reconstructive surgery had lower knee flexion angles and internal knee extensor moments during weight acceptance than uninjured subjects, similar to the adaptation that ACL deficient subjects exhibit following injury [7,8]. Subjects with nearly normal quadriceps strength following reconstruction appeared to have sufficient eccentric quadriceps control during weight acceptance to resume normal knee movement patterns.

The weak-ACLR group's walking and jogging movement patterns strongly resembled those of the unstable ACL-D group, despite the surgical restoration of knee stability. Both the weak and strong subjects had undergone successful ACL reconstruction as manifested by no symptoms of instability, full passive and active knee range of motion, and no joint effusion or pain. The

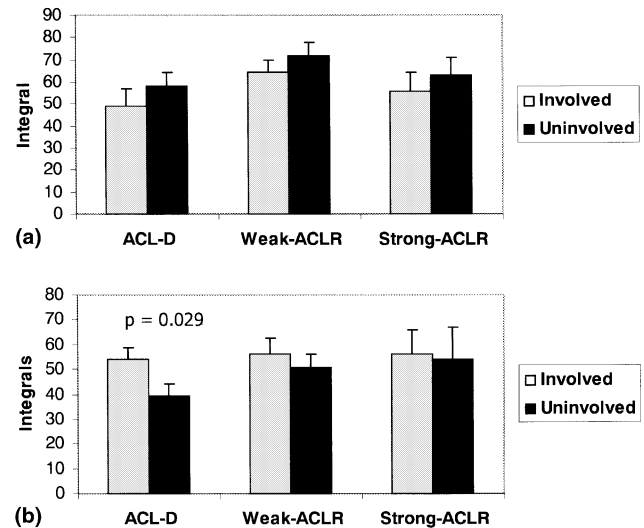


Fig. 5. Muscle activity during jogging integrated from 100 ms prior to initial contact through peak knee flexion. Data represent the vastus lateralis (a) and lateral hamstrings (b) for the ACL-D, weak-ACLR and strong-ACLR groups.

reduced knee angles and moments characteristic of the gait of ACL deficient individuals with instability, however, were present in the weak-ACLR group. These data support the contention that quadriceps weakness contributes to, but does not fully explain, the gait adaptations in individuals with ACL injury.

The results of this study are consistent with the literature and help to complete the picture developed by other investigators. DeVita et al. [13] reported that despite 6 months of rehabilitation, subjects following ACL reconstruction continued to have diminished knee moments during early stance compared to controls. Timoney and colleagues [29] also reported that following reconstruction, subjects exhibited diminished knee moments during weight acceptance. Quadriceps strength testing was not performed in these studies so the relationship between quadriceps strength and decreased knee moment cannot be ascertained. Hurwitz et al. [21] reported that the internal knee extension moment during jogging was significantly correlated with isokinetic knee extension strength after ACL reconstruction. Snyder-Mackler et al. [20] reported that isokinetic quadriceps indices were significantly correlated with flexion excursion during the early stance phase of walking in subjects following ACL reconstruction. Our data confirm the findings of all these authors, and provide additional evidence that quadriceps weakness contributes to the characteristic movement patterns in these patients during both walking and jogging.

The reduced knee extensor moments during weight acceptance in the weak-ACLR group are not attributable to reduced quadriceps activation, as vastus lateralis

muscle activities were not different in the involved and uninvolved sides. The full quadriceps activation found in our weak-ACLR subjects is consistent with the findings of Bulgerhoni et al. [12], who reported greater quadriceps activation during early stance in people following ACL reconstruction. During jogging, it appears that the ACL-D subjects reduced the knee angles and moments through an increase in hamstring activity. This adaptation has been reported previously [30] as a compensation for knee instability. The higher level of hamstring muscle activation in the ACLD subjects suggests that quadriceps femoris weakness exacerbates the problem of knee instability and requires a different compensation strategy. It is important to note that none of the groups demonstrated a significant reduction in quadriceps activity. The assertion that the reduced knee extension moment that is seen in persons with ACL injury represents an attempt to avoid episodes of instability by decreasing quadriceps activation is therefore unsubstantiated using the current data.

The reduced knee flexion in anterior cruciate ligament deficient and weak reconstructed subjects may have important consequences for long-term joint integrity. Reduced knee flexion interferes with the normal ability of the knee to absorb shock during weight acceptance [31], and may be associated with early degenerative joint changes. A stiffer knee, which does not flex sufficiently, is often observed following ACL rupture or reconstruction [8,12,13]. Our data provide evidence that quadriceps strength deficits may be a factor influencing this reduction in knee flexion, and therefore might be responsible for the increased risk of early degenerative changes. Evidence exists to suggest that the rate of loading, which was not tested in this study, on articular cartilage may be more detrimental to the joint surfaces than the magnitude of the joint contact forces [32]. Quadriceps weakness has recently been associated with an increased knee joint loading rate in females [33], which bodes poorly for the integrity of the joint's contact surfaces.

If quadriceps femoris strength deficits are implicated in the development of early degenerative joint arthritis, then return to high-level activity with weak quadriceps after ACL reconstruction appears ill advised. "Return to play" guidelines are not clearly defined following anterior cruciate ligament reconstruction. While quadriceps femoris strength is often used as a marker for rehabilitation progress, return to play criteria vary widely [34,35]. Shelbourne and Foulk [34] allow patients to return to sport specific training with quadriceps index of 65%, but not to return to full competition until the quadriceps index is greater than 80%. Others have used a criterion measure of a quadriceps index greater than 90% for return to sport [35]. More recently, DeCarlo and colleagues [36] have advocated a practice that allows patients to return to activity when their knees have full range of motion, no effusion and the "knees can be rehabilitated to feeling

normal to them with everyday activities". All of the subjects in our study met these criteria: they had self-ratings of knee function that were high, no effusion and normal ROM. However, the group with a quadriceps index of less than 80% had walking and jogging patterns that were similar to patients with ACL deficiency, and these adaptations may be detrimental to joint integrity which suggests that full ROM, no effusion, and high self-rating scores of function may be insufficient criterion to allow for "return to play".

Clearly, knee instability affects gait patterns [7,11]. Our data support the hypothesis that quadriceps femoris strength deficits also contribute significantly to an altered gait pattern after anterior cruciate ligament reconstruction. Strong quadriceps appear to be necessary to allow for a return of normal joint kinematics, kinetics and muscle activities following ACL reconstruction.

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